GEBZE TECHNICAL UNIVERSITY
ELECTRONICS ENGINEERING DEPARTMENT

ELM 366 COMMUNICATIONS LABORATORY
REPORT 2

EXPERIMENT 2: SSB And FM Modulation And Demodulation

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1 Introduction

In this laboratory experiment, the focus was on exploring the principles of Single Sideband (SSB) Modulation & Demodulation and Frequency Modulation (FM) & Demodulation in communication systems. We commenced by generating SSB signals and employing demodulation techniques to recover the embedded message. This hands-on approach provided valuable insights into the bandwidth efficiency achieved by SSB modulation.

Additionally, we delved into the intricacies of Frequency Modulation (FM) and its demodulation process. Through practical exercises, we manipulated the frequency of a carrier signal to encode information, and subsequently, we demodulated the signal to retrieve the original message. This process highlighted the reliability of FM in transmitting information over varying bandwidths and its robustness in recovering the message signal.

2 Theoretical Background

2.1 Single Side-band Modulation and demodulation

Single Sideband (SSB) modulation is a technique used in telecommunications to transmit radio signals. It involves suppressing one of the sidebands of an amplitude-modulated (AM) signal while keeping the other sideband and the carrier wave as shown in Figure 1.[1] This results in a more efficient use of bandwidth compared to traditional AM modulation, where both sidebands are transmitted.

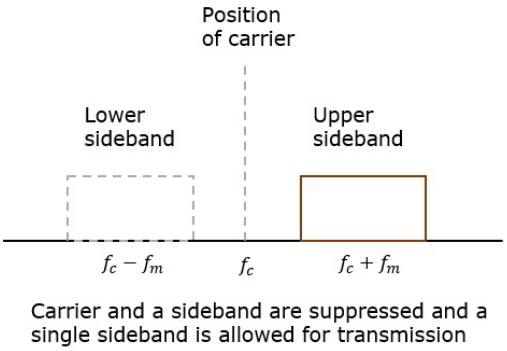


Figure 1

The modulated signal can be defined as follow:

$$if\ m(t)\ is\ the\ message\ signal$$

$$y_{SSB}(t) = m(t)\cos(2\pi f_c t) \mp m_h(t)\sin(2\pi f_c t)$$

$$when\ m_h(t) = m(t) * \frac{1}{\pi t}\ is\ the\ Hilbert\ Tranform\ of\ m(t)$$

$$[Notethat * isconvulation]$$

The block diagram for SSB modulating with Hilbert is given in Figure 2.

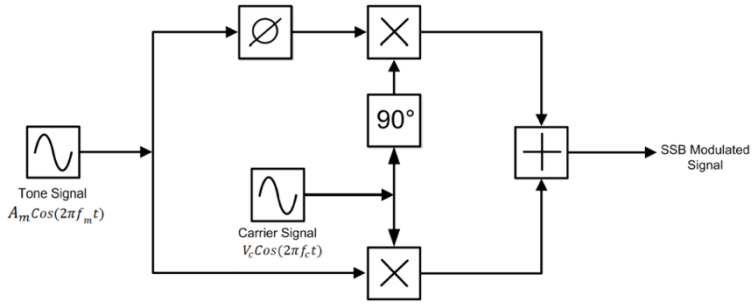


Figure 2 : Modulation of signal with Hilbert Transform

Also the modulated signal can be obtain by modulate the message signal by the carrier than pass it from a bandwidth filter that pass one of the side bands as shown in Figure 3.[2]

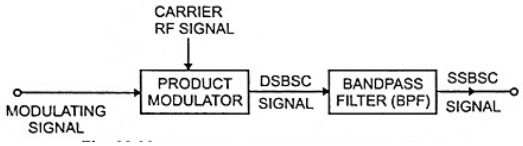


Figure 3: Modulation of SSB signal with BPF

The first step in demodulating an SSB signal is down-conversion. This involves shifting the frequency of the received signal to baseband. Since the carrier frequency (fc) is known accurately, this can be achieved by multiplying the received SSB signal by a local oscillator signal with frequency fc and phase synchronicity. Mathematically, this process can be represented as:

$$y_{SSB\ down\ converted} = y_{SSB}(t)\cos(2\pi f_c t)$$

After down-conversion, the next step is to filter out the high-frequency components introduced during the down-conversion process. Since the original message signal is assumed to be band-limited, it can be recovered by applying a low-pass filter to the down-converted signal.[3] The low-pass filter $(cutoff frequency = f_m)$ removes the high-frequency components, leaving behind the baseband message signal. Mathematically, the output of the low-pass filter can be represented as:

$$m_{demodulated}(t) = LPF(y_{SSB\ down\ converted})$$

The demodulation block diagram draws and shown at Figure 4.

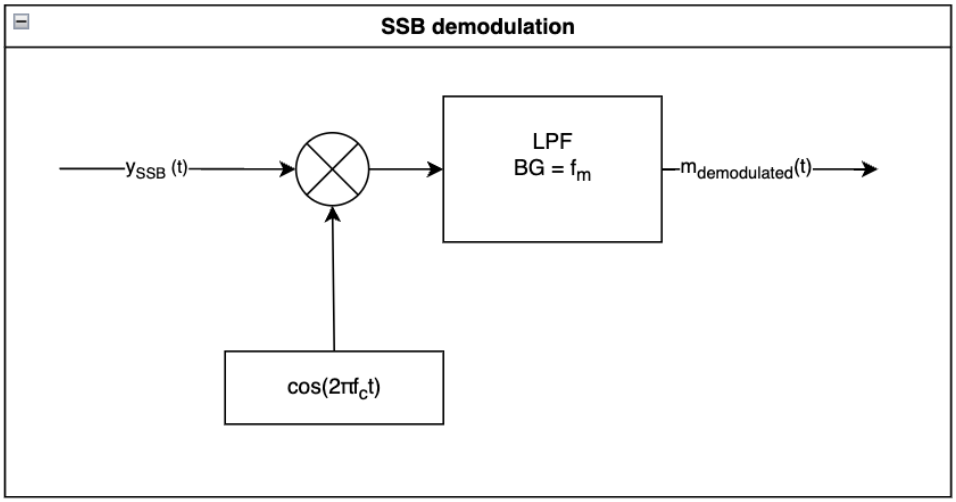


Figure 4: Block diagram of demodulation of SSB

SSB modulation is commonly used in applications such as amateur radio, maritime communications, and military communications. It offers advantages in terms of signal clarity, power efficiency, and reduced interference compared to traditional AM modulation.

At the receiver end, SSB signals require special demodulation techniques to recover the original message. These demodulation methods typically involve reinserting the carrier wave and one of the sidebands to reconstruct the original signal.

2.2 FM modulation and demodulation

Frequency Modulation (FM) is a modulation technique used in telecommunications to encode information in the frequency of a carrier wave. In FM modulation, the frequency of the carrier signal is varied in accordance with the amplitude of the modulating signal (message signal). This results in a signal with a constant amplitude but varying frequency.

When $m(t)$ is the message signal and the carrier is $A_c \cos(2\pi f_c t)$

$$y_{FM}(t) = A_c \cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\, d\tau\right.$$

Where:

- $y_{FM}(t)$ is the FM modulated signal.
- A_c is the amplitude of the carrier wave.
- f_c is the carrier frequency.
- k_f is the frequency sensitivity constant.
- $m(t)$ is the modulating (message) signal.

The frequency deviation (Δf) is the maximum amount by which the carrier frequency is allowed to deviate from its unmodulated value. It is determined by the amplitude of the modulating signal and the frequency sensitivity constant (k_f).

And the modulation index is defined as following:

$$\beta = \frac{\Delta f}{f_m}$$

The formula to calculate the bandwidth (B) of an FM system is given by Carson's rule:

$$B = 2(\Delta f + fm) = 2(\beta + 1)f_m$$

If the message signal is sinusoidal so $m(t) = A_m \cos(2\pi f_m t)$

The FM signal can be described by the following equation:

$$y_{FM}(t) = A_c \cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\, d\tau = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t))\right)$$

When

$$\beta = \frac{\Delta f}{f_m} = \frac{k_f A_m}{fm}$$

So

$$y_{FM}(t) = A_c \sum_{-\infty}^{\infty} (J_n(\beta) \cos(2\pi(f_c + n\, f_m)t))$$

An example sinusoidal message modulation is given as following in Figure 5.

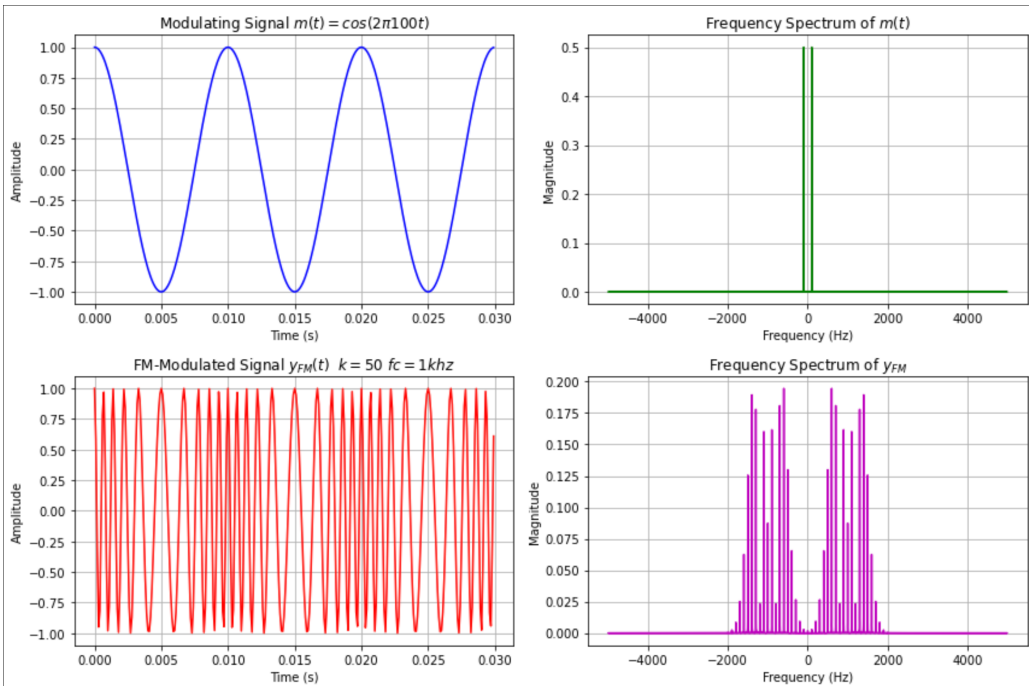


Figure 5

Demodulation of FM Signals using Comparator, Zero-Crossing Detector, and LPF:

Comparator and Zero-Crossing Detector:

- The FM signal is first applied to a frequency-to-voltage converter or a frequency discriminator.
- The output of the frequency discriminator is then fed into both a comparator and a zero-crossing detector circuit.
- The comparator compares the voltage variations from the discriminator with a reference voltage.
- When the voltage exceeds the reference voltage, the comparator generates pulses.
- Simultaneously, the zero-crossing detector produces pulses each time the FM signal crosses zero amplitude.
- Both sets of pulses carry information about the frequency variations in the original FM signal.

Low-Pass Filter (LPF):

- The output pulses from the comparator and zero-crossing detector are passed through a low-pass filter.
- The LPF removes high-frequency components, leaving behind the original message signal.
- Since the output pulses from the comparator and zero-crossing detector represent the frequency variations of the FM signal, the LPF effectively extracts the message signal from these variations.

The block diagram of FM demodulation [4] given in Figure 6 and an example of it at Figure 7.

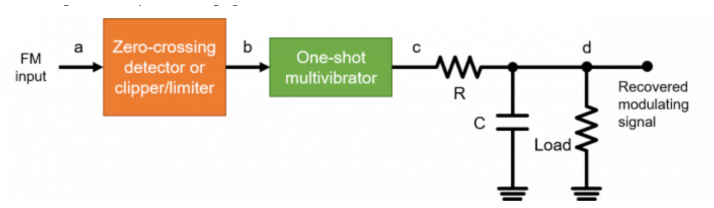
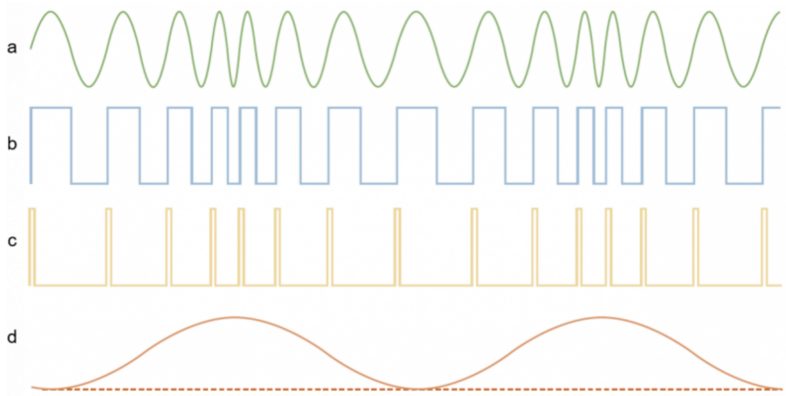


Figure 6



1. I Pulse-Averaging Discriminator's Waveforms: (a) FM input, (b) Output of zero-crossing detector, © Output of one shot, (d) Output of discriminator (original modulating signal).

Figure 7

3 Experiment

3.1 Generating Single Side Band (SSB) Signal

3.1.1 Time domain observation

To generate an SSB signal the connections made as given in Figure 5. The following connections and adjustments were made using Tetra BB control interface:

The carrier frequency (fc) was set to 100kHz, while the message frequency (fm) was set to 2 kHz with a peak-to-peak amplitude of Vpp = 4 V for the message signal m(t). Additionally, the phase shifter was adjusted to 90°, and the input signal frequency in the phase shifter block was set. The gain parameters (g and G) of the adder block were both set to 1. Finally, adjustments were made on the oscilloscope panel, with Channel 0 Voltage Range set to 2V/div, Channel 1 Voltage Range to 2V/div, Horizontal Time base to 100µs/div, and Trigger set to Immediate.

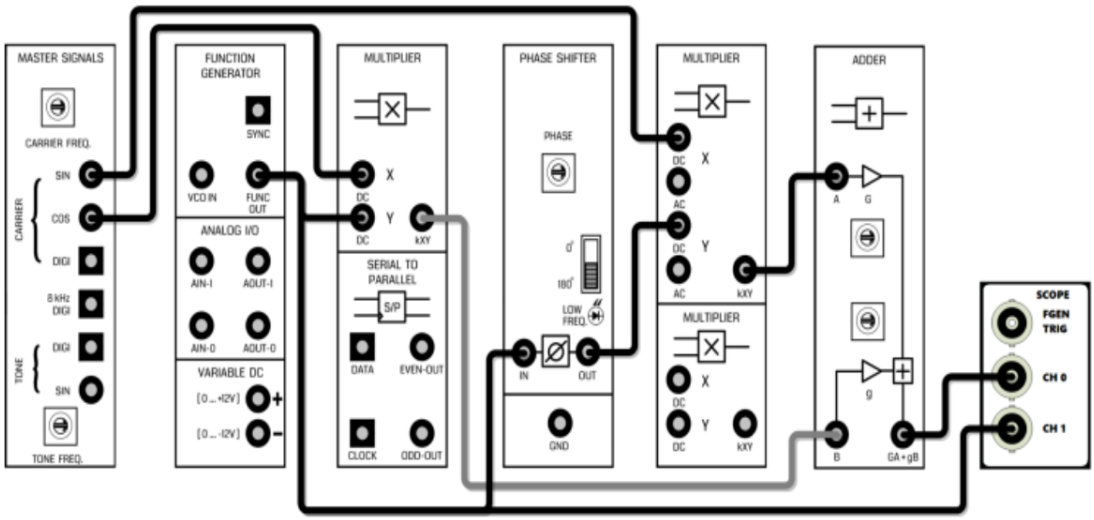


Figure 8: Wiring of TETRA-BB for generating SSB signal

As seen in Figure 5, the message signal is multiplied by the carrier (carrier sin), and the message signal is then shifted 90 degrees using a phase shifter. Afterward, it is multiplied with the carrier phase-shifted version (cos). Finally, the two signals are added together, resulting in the successful generation of the SSB signal. This process can be shown as the following:

$$m(t) = A_m \sin(2\pi f_m t)$$

$$c(t) = A_c \sin(2\pi f_c t)$$

$$y(t) = m(t)c(t) + m_{shifted}(t)c_{shifted}(t)$$

$$y_{SSB}(t) = A_c\, A_m \sin(2\pi f_m t)\sin(2\pi f_c t) + A_c\, A_m \cos(2\pi f_m t)\cos(2\pi f_c t)$$

$$y_{SSB}(t) = A_c A_m \cos\left(2\pi(f_c + f_m)t\right))$$

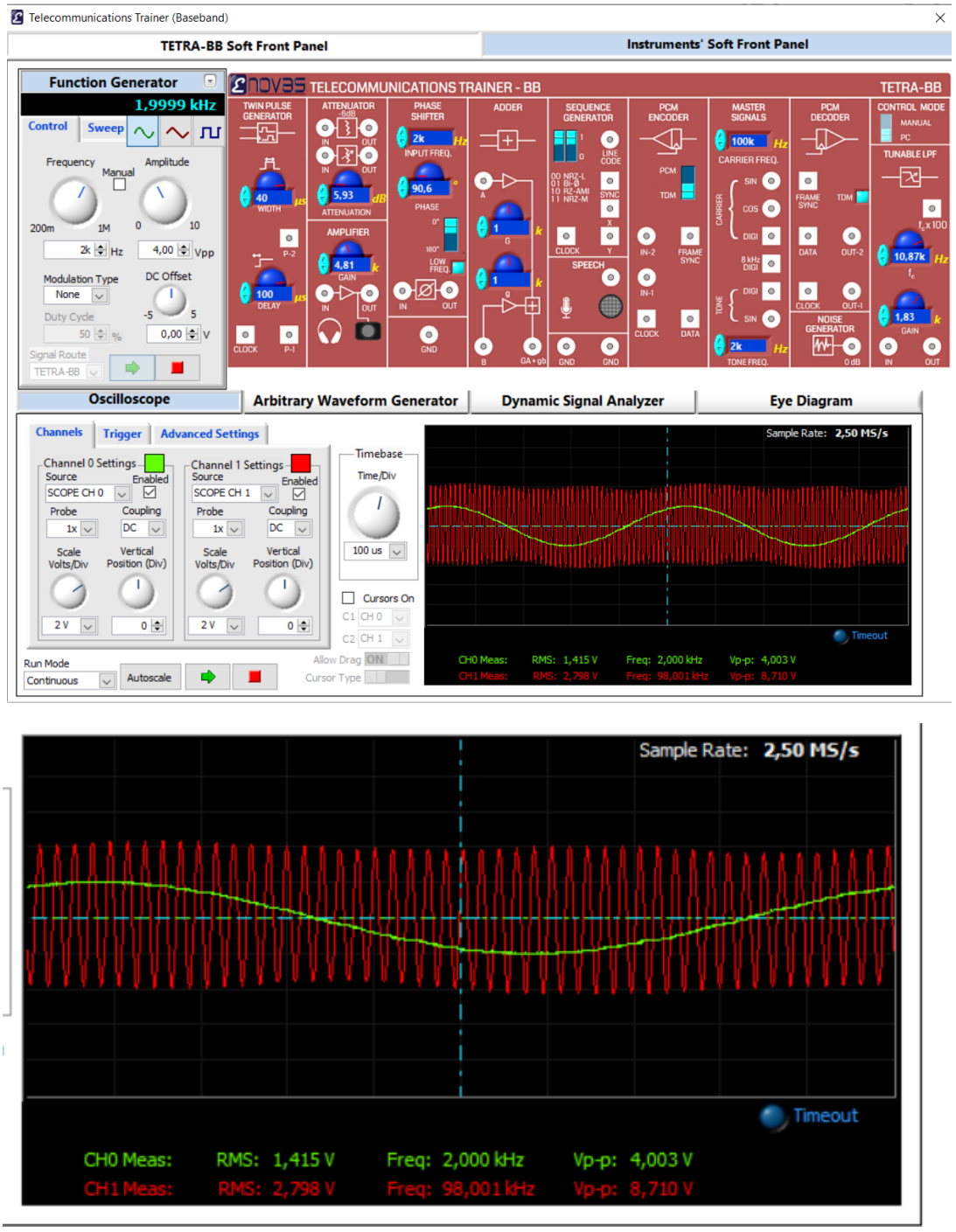


Figure 9: SSB signal

In Figure 6 the CH0 is connected to the message signal $m(t)$ and the CH1 is connected to the SSB signal $y_{SSB}(t)$ (The connections reversed according to Figure 5).

3.1.2 Frequency spectrum observation

After completing the adjustments on the oscilloscope panel, the oscilloscope was stopped, and the switch was made from the oscilloscope window/mode to the DSA panel. In the DSA panel, the following settings were applied: Channel 0 Center Frequency was set to 100000, Frequency Span to 60000, Mode to RMS, Resolution (lines) to 1600, Weighting to Exponential, Window to 7 Term-B Harris, and the Number of Averages to 5.

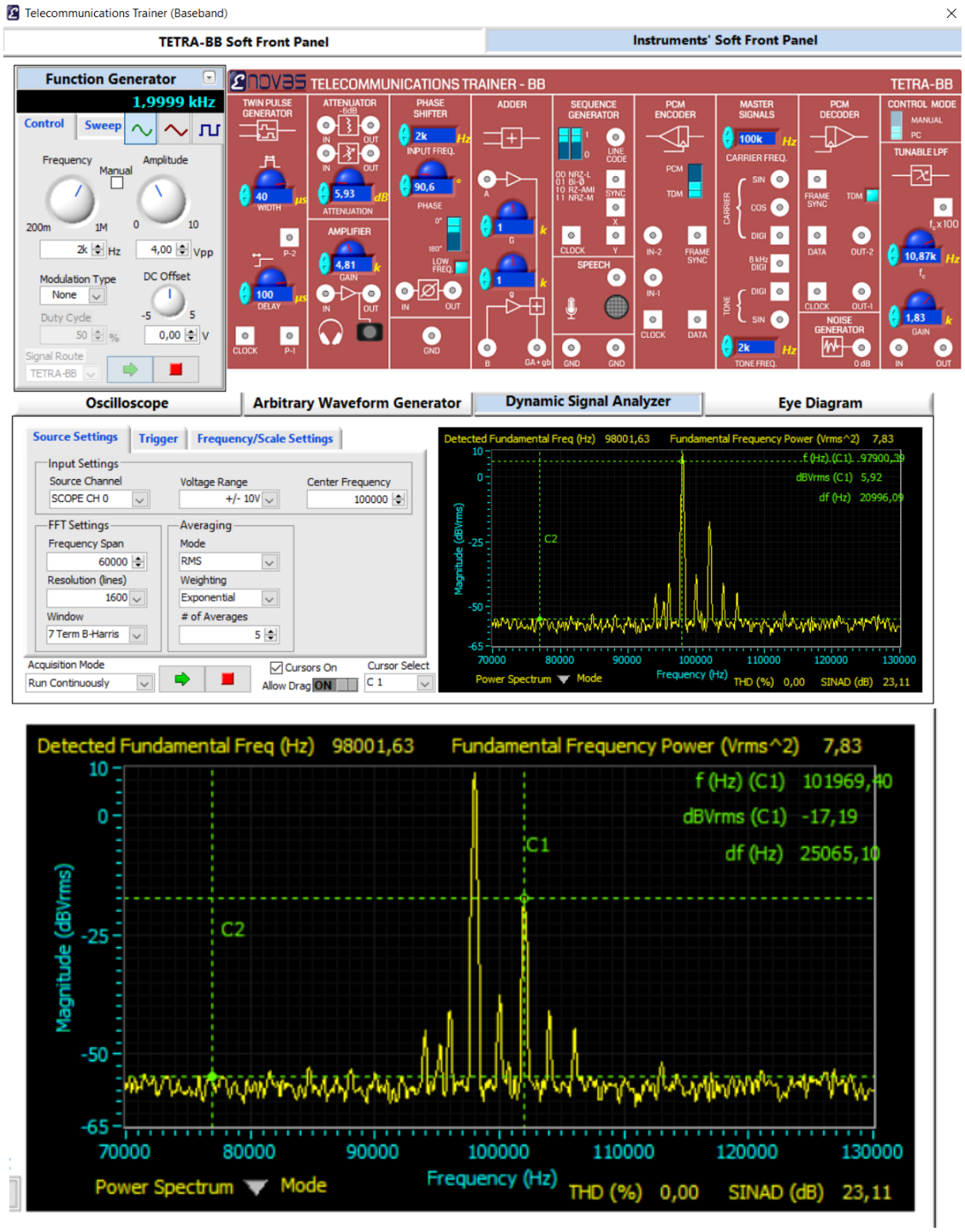


Figure 10: SSB modulated Signal frequency spectrum

The SSB modulated signal frequency spectrum is shown at figure 7. The biggest frequency component is at 98Khz, with magnitude of 5.9 dBVrms because the modulated signal is suppressed lower band SSB modulated signal. $f = fc - fm = 100 - 2 = 98\,kHz$.

The next biggest component after the first one has a magnitude is -17 dBVrms (at 102 kHz) this can be assumed as noise. So the main component is at 98 kHz and the signal is modulated success.

3.2 Demodulating SSB signals

To implement the process of DSB demodulation and required TETRA-BB connections were depicted in Figure 8. The gain of the tunable low pass filter was initially set to 1. Then, the cut-off frequency of the tunable low pass filter was adjusted to 5kHz.

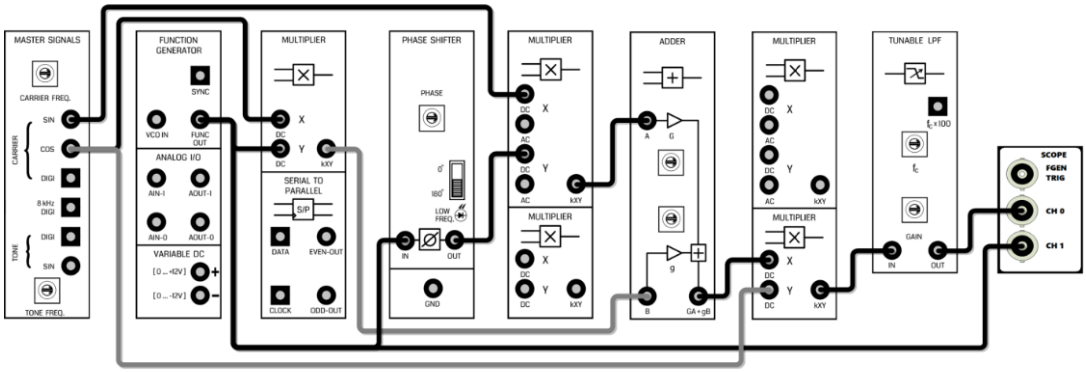


Figure 11: Wiring of TETRA-BB for demodulating of SSB signal

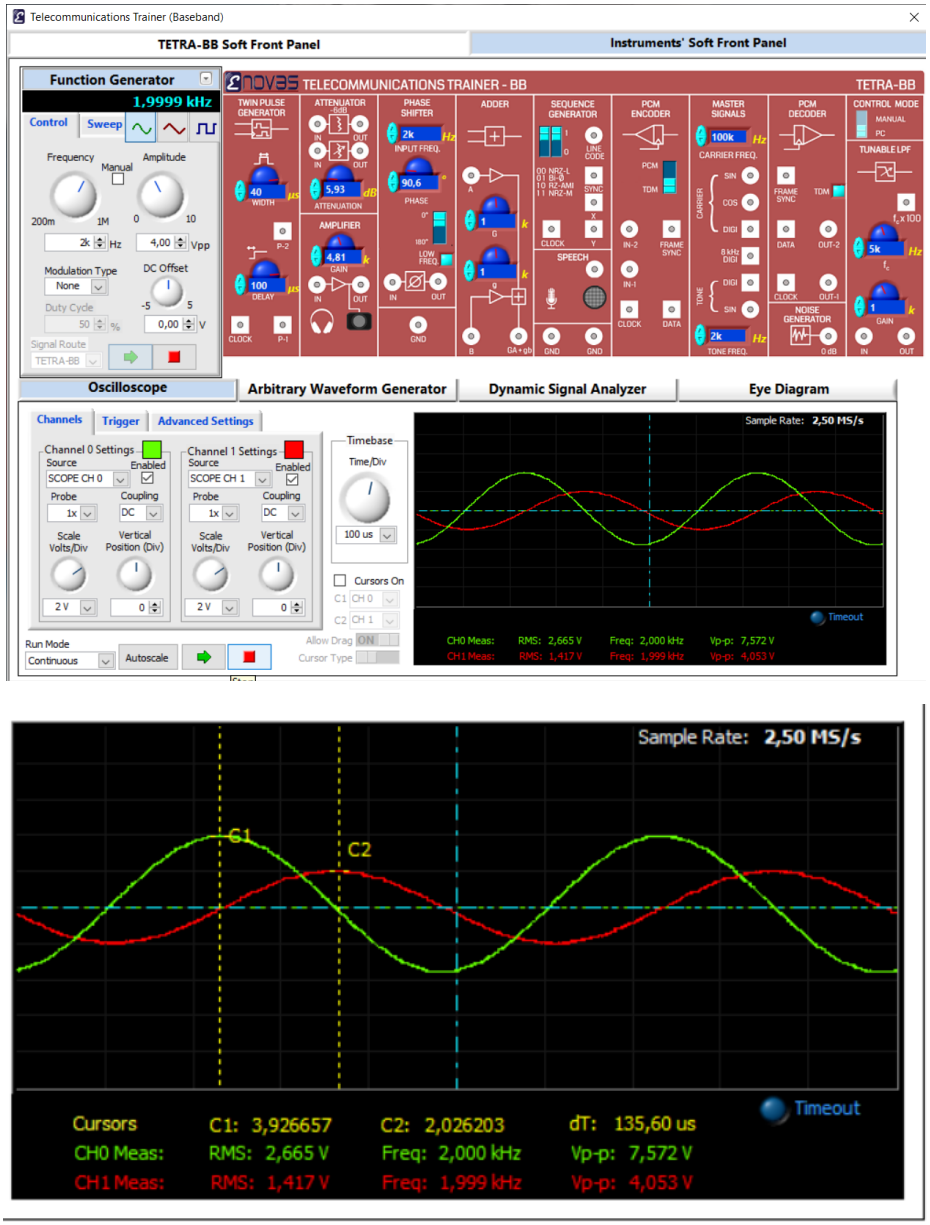


Figure 12: Demodulated SSB signal

By multiplying the SSB signal with the carrier and passing it through the low-pass filter, the demodulated signal (CH1 Red one) is obtained, as shown in Figure 9. The frequency of both the message signal and the demodulated signal is observed to be 2 kHz from Figure 9. The SSB signal is successfully demodulated. It's worth noting that the amplitude of the demodulated signal is half that of the message signal due to the demodulation process, but this is not an issue.

3.3 Generating an FM signal

3.3.1 FM modulation for DIGI as message signal time domain observation

This experiment provided guidance for implementing the FM modulator. The block diagram and required TETRA-BB connections, as illustrated in Figure 10, were utilized. The peak-to-peak amplitude of the sinus signal in the Function Generator has been adjusted to Vpp = 4 V and 50 kHz, and the digital tone signal frequency (fm) has been set to 2 kHz for the message signal m(t) in the Master Signal block. Additionally, the "Modulation Type" has been set to FM in the function generator, and the gain (g) of the adder has been adjusted to 0.3. Finally, the voltage ranges and time base on the oscilloscope panel have been adjusted to clearly visualize the obtained signals.

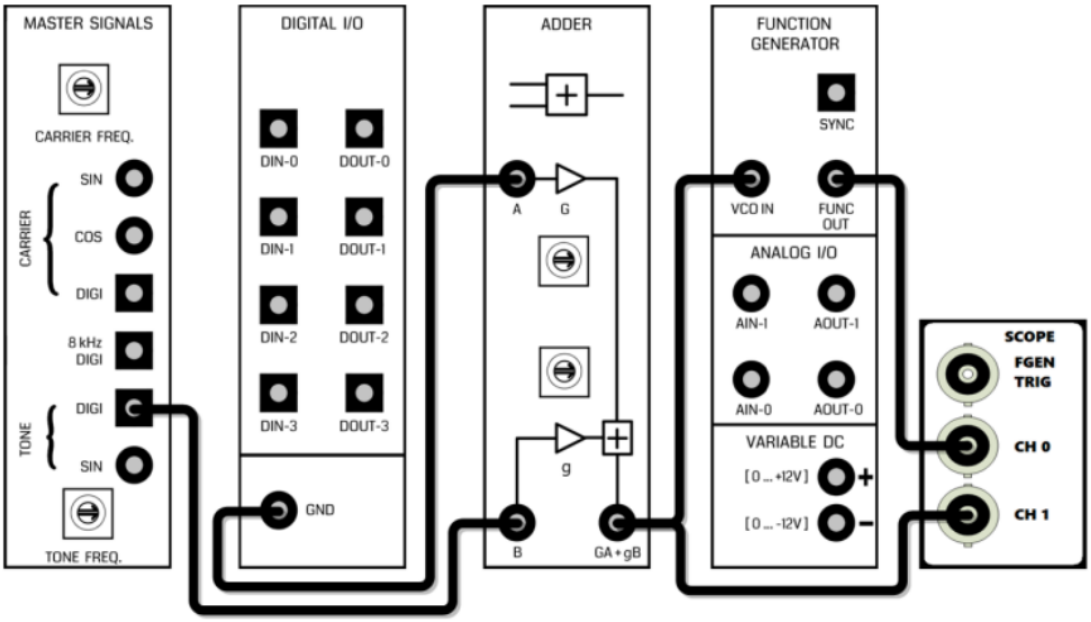


Figure 13: FM signal with DIGI as message signal

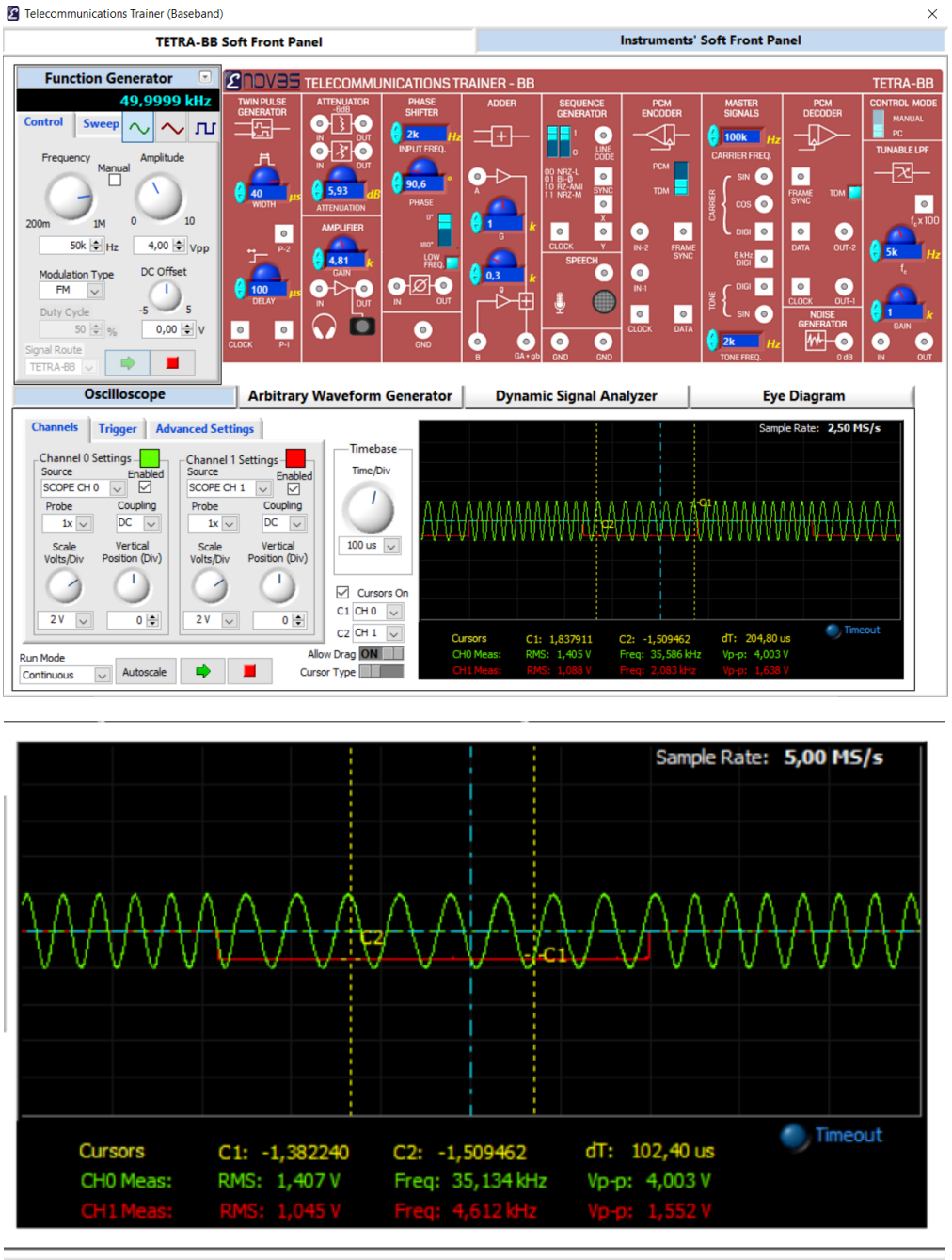


Figure 14: FM signal for DIGI input

In FM modulation, the frequency of the FM signal varies in accordance with the amplitude of the message signal, $m(t)$. Therefore, when the amplitude of the message signal decreases, the frequency of the FM signal also decreases. Consequently, the period of the FM signal increases, causing the signal to expand in the time domain. This relationship is clearly visible in Figure 11.

3.3.2 FM modulation for SIN as message signal time domain observation

Following the previous steps, the digital tone signal has been disconnected from the adder block, and instead, the sinus tone from the Master Signals has been connected to the adder block (with g = 1). Now, the modulated signal is being observed for further analysis.

The message signal and the FM is shown in Figure 12. The message signal and the FM are shown in Figure 12. As with the digital signal used as the message signal, the frequency of the carrier varies due to the amplitude of the message signal. However, because the message signal is sinusoidal, the deviation of the frequency is smoother compared to the digital signal.

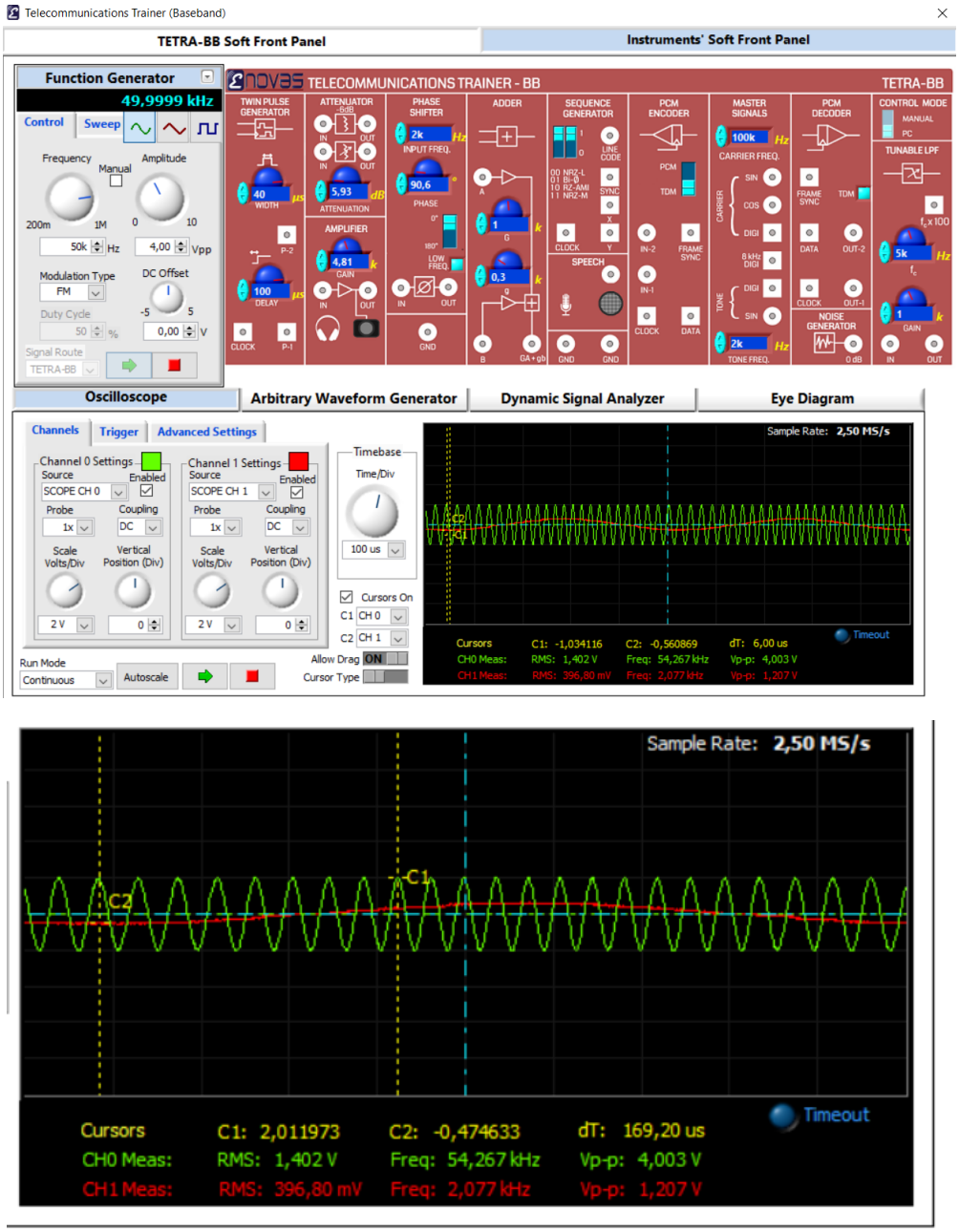


Figure 15: FM signal for SIN as message signal

3.3.3 FM signal frequency spectrum for DIGI input

The oscilloscope operation was halted, and a transition was made from the oscilloscope window/mode to the DSA panel. Subsequently, the specified settings for the DSA were implemented: Channel 1's center frequency was adjusted to 50000, with a frequency span of 100000. The mode was set to RMS, employing a resolution of 1600 lines and an exponential weighting. Additionally, a window setting of 7 Term-B Harris was applied, along with an averaging number of 5 for enhanced analysis.

The FM signal frequency spectrum is shown at Figure 13. The magnitude of the components before 25 kHz and after 55 kHz is less than -20 db, so it can be accepted as noise. And theBG is near 30khz.



Figure 16: FM signal frequency spectrum for DIGI input

3.3.4 FM signal frequency spectrum for SIN input

Following the previous steps, the digital tone signal has been disconnected from the adder block, and instead, the sinus tone from the Master Signals has been connected to the adder block (with g = 1).

Now, the FM signal frequency spectrum is being observed for further analysis as shown in figure 14. The magnitude of the components before 40 kHz and after 60 kHz is less than -20 db, so it can be accepted as noise. And the BG is near 20kHz.

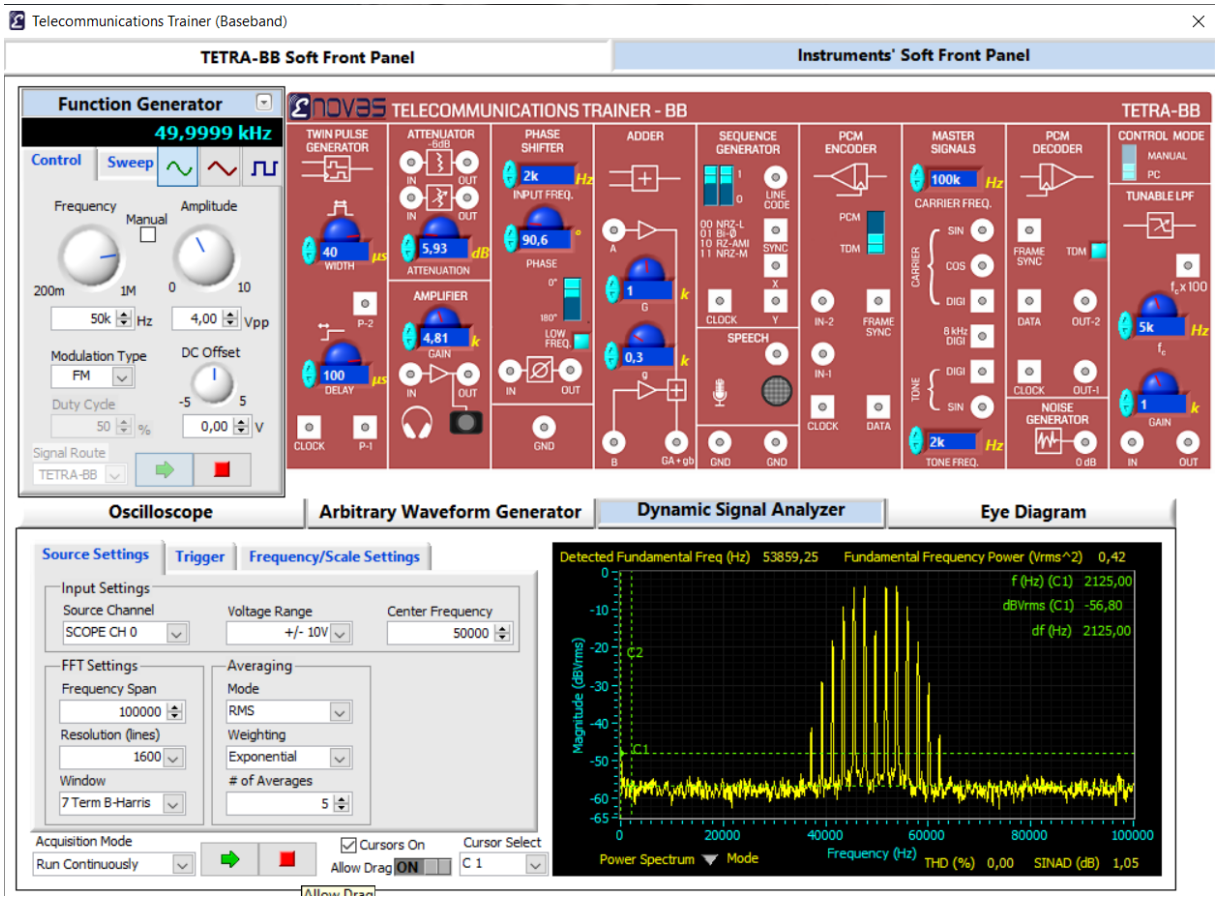


Figure 17: FM signal frequency spectrum for SIN input

3.4 FM demodulation

For FM demodulation, the wiring was arranged according to the configuration depicted in Figure 15. Subsequently, adjustments were made as follows: g was set to 0.3, and fm was fine-tuned to 5kHz. The cut-off frequency of the tunable low pass filter was precisely adjusted to regenerate the message signal at the output of the filter, achieving a frequency of 7.89kHz with a gain of 2. Additionally, other settings were tailored accordingly: the Twin Pulse Generator's width was precisely configured to 5µs, with a delay of 405µs, while the amplifier gain was meticulously set to 7 to ensure optimal performance.

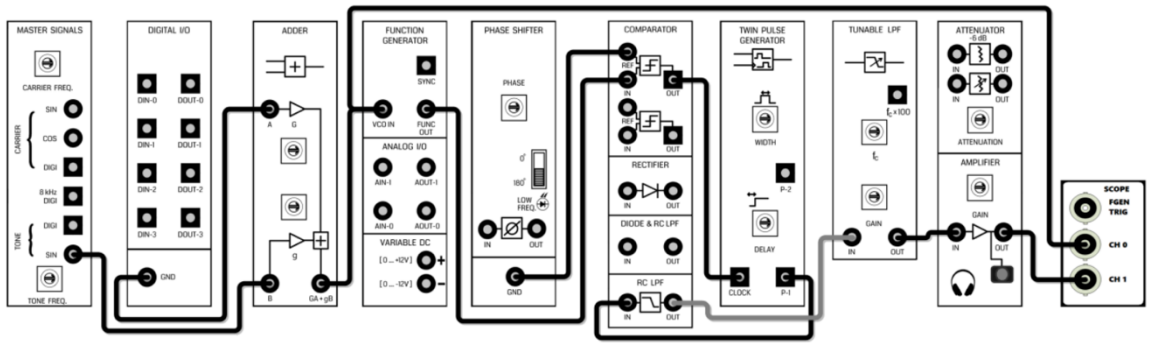


Figure 18: Wiring of TETRA-BB for FM demodulation

The result of demodulation is obtained as shown in Figure 16 with dc coupling. The demodulated signal has an dc component. To make analyze better to observe the Dc component is ignored by using AC coupling as shown at Figure 17.

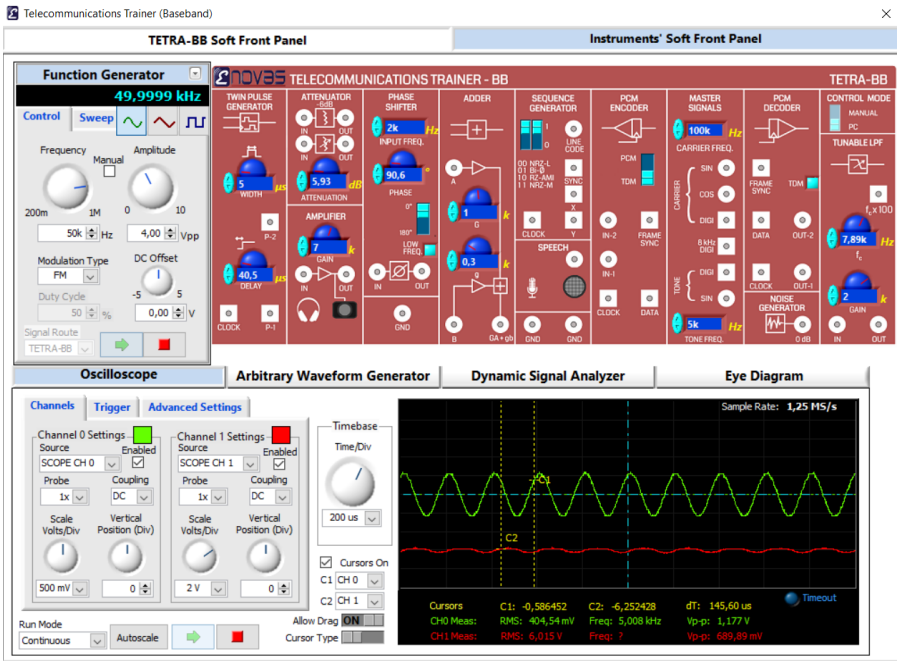


Figure 19: FM signal demodulation DC coupling

As shown at Figure 17, the message signal is obtained success with by the demodulation of the FM signal.

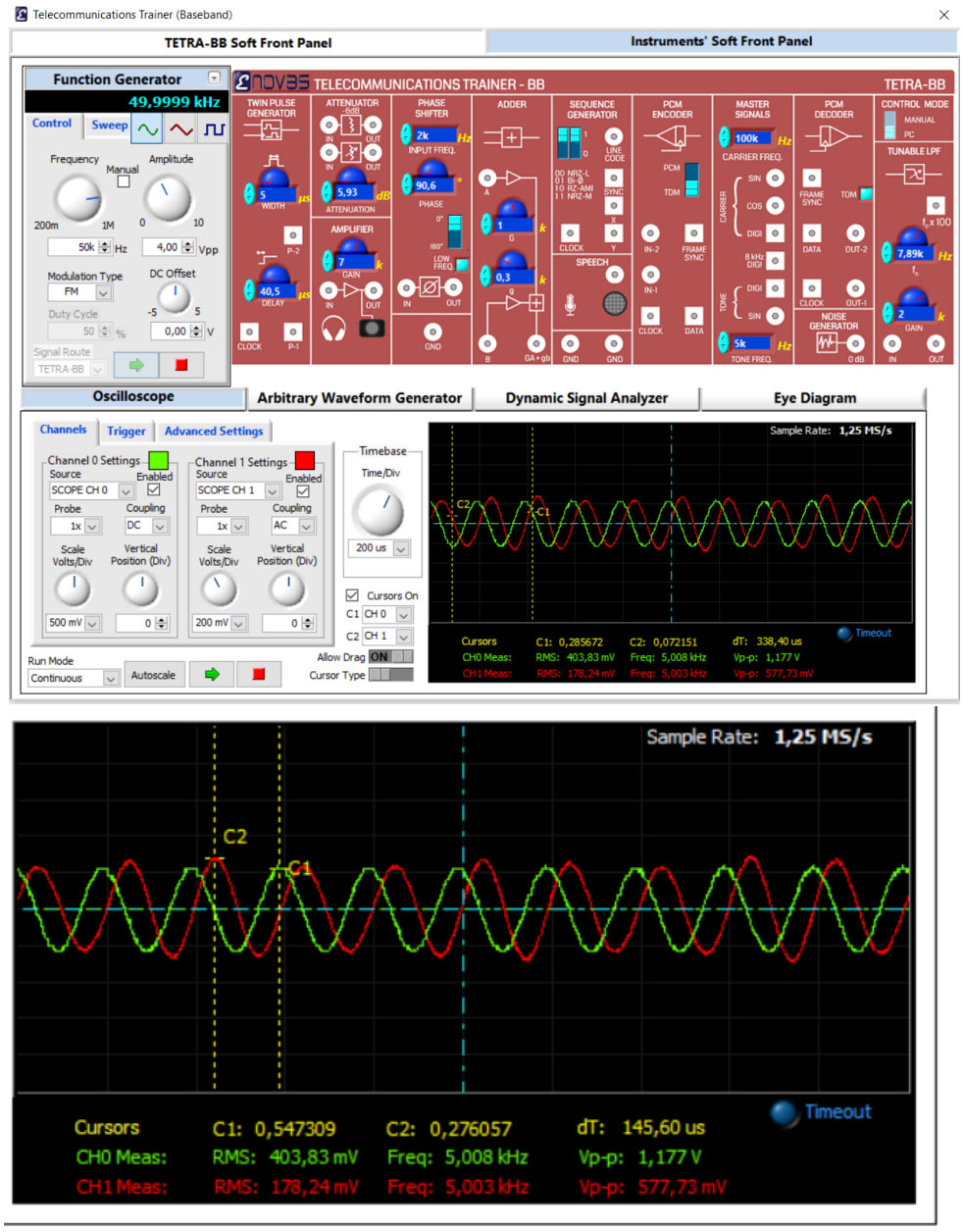


Figure 20: FM signal demodulation AC coupling

4 Conclusion

In this lab, knowledge regarding Single Side Band (SSB) and Frequency Modulation (FM) communication techniques was acquired and experiments were conducted to explore their characteristics. It was observed that SSB communication necessitates less bandwidth compared to FM, indicating its efficiency in utilizing available spectrum. Conversely, FM requires a substantially broader bandwidth, albeit offering enhanced communication capabilities. Although the FM demodulator is more intricate than SSB, its advantages in achieving superior communication quality were evident.

Moreover, it is noteworthy that FM modulation facilitates the transmission of high-quality audio signals, rendering it extensively utilized in applications such as radio broadcasting and music transmission. Despite its complexity, FM modulation offers benefits in terms of signal clarity and resistance to noise.

5 References

[1]https://www.tutorialspoint.com/principles_of_communication/principles_of_communication_sideband_modulation.htm

[2] https://www.eeeguide.com/single-sideband-modulation-ssb/

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